

Project Concept

Design & Implementation of a Secure Multi-Floor Enterprise Network Infrastructure Using Cisco Technologies

This project demonstrates the practical implementation of a secure enterprise network environment using Cisco networking technologies. The network is designed to simulate a real-world multi-floor organizational infrastructure where different departments are logically separated while maintaining controlled communication and enterprise-level security.

The project focuses on network segmentation, routing, branch connectivity, security implementation, and secure remote management to create a stable and efficient enterprise network.

Summarized Scenario

A secure multi-floor enterprise network using Cisco Technologies must be designed with VLAN segmentation, Inter-VLAN routing, Branch office connectivity through dynamic routing, SSH remote management and basic network security.

Project Requirements

The following requirements were provided in the project scenario and have been implemented in this network design.

1. Design a Multi-Floor Enterprise Network

The network must represent a multi-floor organization with multiple departments connected through an enterprise topology.

2. Department Segmentation Using VLANs

Different departments must be logically separated using:

Virtual Local Area Networks (VLANs)

to improve:

- Security
 - Traffic control
 - Department isolation
 - Broadcast management
-

3. Configure Inter-VLAN Routing

Different VLANs must communicate through:

Inter-VLAN Routing

using controlled communication.

This was implemented using:

Router-on-a-Stick

4. Controlled Communication Between Departments

The network must enforce communication rules where:

- Same department devices can communicate
- Unauthorized department access is restricted
- Management department can access all departments

This requirement was implemented using:

Access Control Lists (ACL)

5. Branch Office Connectivity

The branch office must communicate with the head office.

A dedicated WAN connection was required between:

Main Office



Branch Office

6. Dynamic Routing

Branch communication must be established through:

Dynamic Routing

instead of static routes.

This project uses:

EIGRP (Enhanced Interior Gateway Routing Protocol)

7. Secure Remote Management

The network must support:

Secure Remote Access

for administrators.

This was implemented using:

SSH (Secure Shell)

instead of Telnet.

8. Basic Network Security

The enterprise network must include security mechanisms such as:

- Firewall protection
 - ACL security
 - Password encryption
 - Login warning banner
 - Secure management access
 - Disabled unused ports
-

9. Wireless Management Access

Management users should be able to access the network wirelessly for easier administrative mobility.

This was implemented using:

Wireless Access Point

for:

- Management Laptop
 - Management Phone
-

10. Enterprise Network Stability & Redundancy

The network should provide improved bandwidth and redundancy between switches.

This was implemented using:

EtherChannel (LACP)

Tools & Technologies Used

Software Used

1. Cisco Packet Tracer

Used for:

- Network design
 - Device configuration
 - Simulation
 - Testing
 - Troubleshooting
-

Networking Devices Used

1. PT Cloud

Used to simulate:

Internet / WAN Environment

2. ISP Router

Purpose:

Internet Service Provider Simulation

3. Cisco ASA 5505 Firewall

Purpose:

- External/Internal traffic protection
 - Security boundary creation
 - Firewall implementation
-

4. Main Router

Purpose:

- Inter-VLAN Routing
 - Dynamic Routing
 - ACL implementation
 - Branch communication
-

5. Branch Router

Purpose:

- Branch office communication
 - Dynamic routing participation
-

6. Core Switch

Purpose:

- Central switching
 - VLAN distribution
 - Trunking
 - EtherChannel aggregation
-

7. Access Switches

Used:

SW1

SW2

SW3

Branch Switch

Purpose:

- End device connectivity
 - Department segmentation
-

8. Wireless Access Point

Used for:

Wireless Management Access

9. End Devices

Used devices:

- PCs
 - Laptop
 - Smartphone
 - Servers
-

Networking Technologies Used

1. VLAN (Virtual Local Area Network)

Used for:

Department Separation

2. Inter-VLAN Routing

Used for:

Communication Between VLANs

3. Router-on-a-Stick

Used for:

Inter-VLAN Routing

through a single router interface.

4. Trunking (802.1Q)

Used for:

Carrying Multiple VLAN Traffic

between switches and router.

5. EtherChannel (LACP)

Used for:

Bandwidth Increase

Redundancy

Load Balancing

6. EIGRP Dynamic Routing

Used for:

Branch Office Communication

7. ACL (Access Control List)

Used for:

Traffic Filtering

Department Security

Controlled Access

8. SSH (Secure Shell)

Used for:

Secure Remote Administration

9. Firewall Security

Used for:

Enterprise Perimeter Security

10. WPA2 Wireless Security

Used for:

Secure Wireless Authentication

11. Password Encryption

Used for:

Credential Protection

12. Login Banner (MOTD)

Used for:

Unauthorized Access Warning

13. ICMP Inspection

Used for:

Ping Communication Through Firewall

ACTUAL NETWORK TOPOLOGY + DEVICE ROLES + PHYSICAL CONNECTIONS + ISP ROUTER + ASA FIREWALL

(11 - 22)

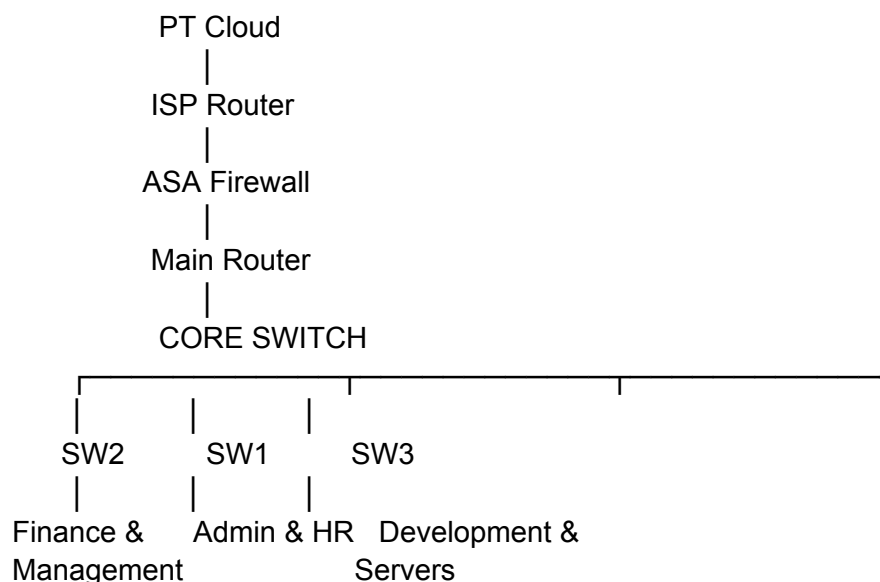
Actual Network Topology

The network was designed as a **multi-floor enterprise network** consisting of multiple departments connected through a centralized architecture.

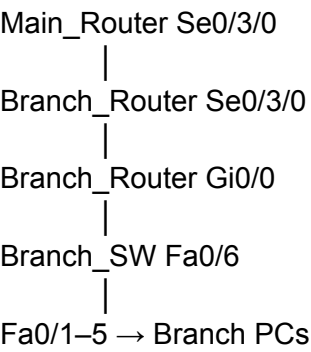
The network includes:

- Head Office
- Branch Office
- Department Segmentation
- Wireless Management
- Enterprise Security

Head Office Topology



Branch Office Topology



Real Physical Connection Mapping

This section contains the **actual implemented cable connections** of the final project.

Internet Side Connections

Device	Interface	Connected To	Interface
PT Cloud	Serial	ISP Router	Se0/3/0
ISP Router	G0/0	ASA Firewall	Eth0/0

Firewall to Main Router

Device	Interface	Connected To	Interface
ASA Firewall	Eth0/1	Main Router	G0/0

Core Network Connection

Device	Interface	Connected To	Interface
Main Router	G0/1	CORE_SW	Fa0/1

This link carries:

All VLAN traffic

through:

802.1Q Trunking

EtherChannel Connections

SW1 → CORE_SW

SW1	CORE_SW
Fa0/1	Fa0/5
Fa0/2	Fa0/6

Used for:

Port-Channel 1 (Po1)

SW2 → CORE_SW

SW2	CORE_SW
Fa0/1	Fa0/3
Fa0/2	Fa0/4

Used for:

Port-Channel 2 (Po2)

SW3 → CORE_SW

SW3 CORE_SW

Fa0/1 Fa0/7

Fa0/2 Fa0/8

Used for:

Port-Channel 3 (Po3)

IT Management Connection

Device	Port
IT_ADMIN_PC	CORE_SW Fa0/2

Purpose:

- SSH Access
- Secure Management
- Remote Administration

Assigned VLAN:

VLAN 99

Wireless Management Connection

Device	Port
MGMT_AP	CORE_SW Fa0/9

Assigned VLAN:

VLAN 40

Connected Wireless Devices:

- MGMT_LAPTOP
- MGMT_PHONE

Department Port Mapping

SW1 — Administration + HR

Administration Department

**Device
Ports**

Fa0/3–4

Assigned:

VLAN 10

HR Department

**Device
Ports**

Fa0/5–7

Assigned:

VLAN 20

SW2 — Finance + Management

Finance Department

**Device
Ports**

Fa0/3–4

Assigned:

VLAN 30

Management Department

**Device
Ports**

Fa0/5–6

Assigned:

VLAN 40

SW3 — Development + Server Operations

Development Department

**Device
Ports**

Fa0/3–4

Assigned:

VLAN 50

Server Operations

**Device
Ports**

Fa0/5–6

Assigned:

VLAN 60

Device Roles

ISP Router

Purpose

Simulates:

Internet Service Provider

Functions:

- Simulated internet
 - External network connection
 - Firewall communication
-

ASA Firewall

Purpose

Acts as:

Enterprise Security Boundary

Functions:

- External/Internal separation
- Traffic inspection
- Firewall protection

Main Router

Purpose

Acts as:

Central Routing Device

Functions:

- Inter-VLAN Routing
 - ACL Security
 - EIGRP
 - Branch Communication
-

CORE_SW

Purpose

Acts as:

Core Distribution Switch

Functions:

- VLAN Distribution
 - Trunking
 - EtherChannel
 - Central Switching
-

ISP Router Configuration

Purpose

The ISP Router provides communication between:

PT Cloud ↔ ASA Firewall

and simulates an internet environment.

Step 1 — Enter Configuration Mode

```
enable  
configure terminal
```

Step 2 — Configure Hostname

```
hostname ISP_Router
```

Step 3 — Configure Firewall Facing Interface

```
interface gigabitEthernet0/0  
ip address 10.0.0.1 255.255.255.252  
no shutdown  
exit
```

Explanation

This interface connects:

ISP Router ↔ ASA Firewall

IP Used:

10.0.0.1/30

Step 4 — Configure PT Cloud Interface

```
interface serial0/3/0  
ip address 203.0.113.1 255.255.255.252  
clock rate 64000  
no shutdown  
exit
```

Purpose

Creates WAN simulation.

Step 5 — Save Configuration

copy running-config startup-config

ASA Firewall Configuration

Important Note

Cisco ASA 5505 uses:

VLAN Interfaces

instead of:

Regular Layer-3 Ethernet Interfaces

That is why:

nameif

must be configured under:

interface vlan

Step 1 — Enter Configuration Mode

enable

configure terminal

Step 2 — Configure Hostname

hostname ASA_FIREWALL

Step 3 — Configure Outside Interface

```
interface vlan 2
nameif outside
security-level 0
ip address 10.0.0.2 255.255.255.252
no shutdown
exit
```

Purpose

Connects:

ASA ↔ ISP Router

Security Level:

0 = Untrusted Network

Step 4 — Configure Inside Interface

```
interface vlan 1
nameif inside
security-level 100
ip address 10.0.0.5 255.255.255.252
no shutdown
exit
```

Purpose

Connects:

ASA ↔ Main Router

Security Level:

100 = Trusted Internal Network

Step 5 — Assign Physical Ports

Outside Port

```
interface ethernet0/0
switchport access vlan 2
no shutdown
exit
```

Inside Port

```
interface ethernet0/1
switchport access vlan 1
no shutdown
exit
```

Step 6 — Configure Default Route

```
route outside 0.0.0.0 0.0.0.0 10.0.0.1
```

Purpose:

Send unknown traffic to ISP Router

Step 7 — Enable ICMP Inspection

```
policy-map global_policy
class inspection_default
inspect icmp
exit
```

Purpose:

Allow Ping Through Firewall

Step 8 — Save Configuration

```
write memory
```

MAIN ROUTER + CORE SWITCH + VLAN CREATION + TRUNKING + ROUTER-ON-A-STICK CONFIGURATION

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Main Router Configuration

Purpose of Main Router

The Main Router acts as the **central routing device** of the enterprise network.

It performs the following tasks:

- Inter-VLAN Routing
- Branch Office Communication
- Dynamic Routing
- ACL Security Implementation
- Central Network Routing

The Main Router connects:

Firewall ↔ Internal Network ↔ Branch Office

Physical Connections of Main Router

Interface	Connected Device	Purpose
Gig0/0	ASA Firewall	Internal Security Link
Gig0/1	CORE_SW Fa0/1	VLAN Trunk Link
Se0/3/0	Branch Router Se0/3/0	WAN Communication

Step 1 — Enter Configuration Mode

```
enable
configure terminal
```

Step 2 — Configure Hostname

hostname Main_Router

Purpose

Changes default router name for better administration.

Step 3 — Configure Firewall Facing Interface

Interface Used

GigabitEthernet0/0

Connected To:

ASA Firewall Ethernet0/1

Configuration

```
interface gigabitEthernet0/0
ip address 10.0.0.6 255.255.255.252
no shutdown
exit
```

Purpose

Establishes communication between:

Main Router ↔ Firewall

Step 4 — Configure CORE_SW Link

Interface Used

GigabitEthernet0/1

Connected To:

CORE_SW Fa0/1

Configuration

```
interface gigabitEthernet0/1
no shutdown
exit
```

Purpose

This interface carries:

All VLAN Traffic

through:

802.1Q Trunking

Step 5 — Configure Branch WAN Link

Interface Used

Serial0/3/0

Connected To:

Branch Router Serial0/3/0

Configuration

```
interface serial0/3/0
ip address 172.16.0.1 255.255.255.252
clock rate 64000
no shutdown
exit
```


Purpose

Creates WAN communication between:

Head Office ↔ Branch Office

Step 6 — Configure Router-on-a-Stick

Why Router-on-a-Stick?

Different VLANs cannot communicate directly.

Example:

VLAN 10 ↔ VLAN 20

communication fails unless routing exists.

To solve this problem, the project uses:

Router-on-a-Stick

This technique uses:

One Physical Interface

+

Multiple Subinterfaces

for Inter-VLAN Routing.

Step 7 — Enable Physical Trunk Interface

```
interface gigabitEthernet0/1
no shutdown
exit
```

Purpose

Activates the trunk interface.

VLAN Planning Table

VLAN	Department	Gateway
10	Administration	192.168.10.1
20	HR	192.168.20.1
30	Finance	192.168.30.1
40	Management	192.168.40.1
50	Development	192.168.50.1
60	Server Operations	192.168.60.1
99	IT Management	192.168.99.1

Step 8 — Configure VLAN 10 (Administration)

```
interface gigabitEthernet0/1.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
description Administration
exit
```

Purpose

Acts as:

Default Gateway

for Administration devices.

Step 9 — Configure VLAN 20 (HR)

```
interface gigabitEthernet0/1.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
description HR
exit
```

Step 10 — Configure VLAN 30 (Finance)

```
interface gigabitEthernet0/1.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
description Finance
exit
```

Step 11 — Configure VLAN 40 (Management)

```
interface gigabitEthernet0/1.40
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
description Management
exit
```

Step 12 — Configure VLAN 50 (Development)

```
interface gigabitEthernet0/1.50
encapsulation dot1Q 50
ip address 192.168.50.1 255.255.255.0
description Development
exit
```

Step 13 — Configure VLAN 60 (Server Operations)

```
interface gigabitEthernet0/1.60
encapsulation dot1Q 60
ip address 192.168.60.1 255.255.255.0
description Server_Operations
exit
```

Step 14 — Configure VLAN 99 (IT Management)

```
interface gigabitEthernet0/1.99
encapsulation dot1Q 99
ip address 192.168.99.1 255.255.255.0
description IT_Management
exit
```

Purpose

Gateway for:

IT_ADMIN_PC

used for:

- SSH
 - Remote Management
 - Secure Administration
-

Step 15 — Save Configuration

```
copy running-config startup-config
```

CORE SWITCH CONFIGURATION

Purpose of CORE_SW

The Core Switch acts as the:

Central Distribution Switch

It connects:

- Main Router
 - SW1
 - SW2
 - SW3
 - Wireless Access Point
 - IT_ADMIN_PC
-

Actual CORE_SW Port Mapping

Port	Connected Device
Fa0/1	Main Router G0/1
Fa0/2	IT_ADMIN_PC
Fa0/3–4	SW2 EtherChannel
Fa0/5–6	SW1 EtherChannel
Fa0/7–8	SW3 EtherChannel
Fa0/9	MGMT_AP

Step 16 — Enter Configuration Mode

enable

configure terminal

Step 17 — Create VLANs

VLAN 10 — Administration

vlan 10
name Administration

VLAN 20 — HR

vlan 20
name HR

VLAN 30 — Finance

vlan 30
name Finance

VLAN 40 — Management

vlan 40
name Management

VLAN 50 — Development

vlan 50
name Development

VLAN 60 — Server Operations

vlan 60
name Server_Operations

VLAN 99 — IT Management

vlan 99
name IT_Management

Step 18 — Configure Trunk Port to Main Router

Connection

CORE_SW Fa0/1



Main Router Gi0/1

Configuration

```
interface fa0/1
switchport trunk encapsulation dot1q
switchport mode trunk
no shutdown
exit
```

Purpose

Allows:

Multiple VLAN Traffic

between:

CORE_SW ↔ Main Router

Step 19 — Configure IT_ADMIN_PC Port

```
interface fa0/2
switchport mode access
switchport access vlan 99
no shutdown
exit
```

Purpose

Assigns:

IT_ADMIN_PC

to:

VLAN 99

Step 20 — Configure MGMT_AP Port

Connected Port

Fa0/9

Configuration

```
interface fa0/9
switchport mode access
switchport access vlan 40
no shutdown
exit
```

Purpose

Places:

MGMT_AP

Inside:

Management VLAN

Step 21 — Save Configuration

```
copy running-config startup-config
```

ETHERCHANNEL + SW1 + SW2 + SW3 + ACTUAL PORT ASSIGNMENT CONFIGURATION

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EtherChannel Configuration

What is EtherChannel?

EtherChannel is a technology used to combine multiple physical links into:

One Logical Link

Instead of relying on a single cable, multiple cables work together to improve:

- Bandwidth
- Redundancy
- Reliability
- Load balancing

Why EtherChannel Was Used in This Project

In a multi-floor enterprise network, communication between:

CORE_SW ↔ Access Switches

is critical.

To prevent:

- Network congestion
- Single point failure
- Bandwidth limitation

the project uses:

EtherChannel using LACP

Why LACP?

The project uses:

LACP (Link Aggregation Control Protocol)

because it is:

- Dynamic
- Enterprise standard
- More reliable
- Cisco recommended

Configuration mode used:

mode active

Actual EtherChannel Port Mapping

Switch	Ports	CORE_SW Ports	Port Channel
SW1	Fa0/1–2	Fa0/5–6	Po1
SW2	Fa0/1–2	Fa0/3–4	Po2
SW3	Fa0/1–2	Fa0/7–8	Po3

SW1 Configuration

Purpose

SW1 is responsible for:

Administration Department
HR Department

Step 1 — Enter Configuration Mode

```
enable
configure terminal
```

Step 2 — Create VLANs

Administration VLAN

```
vlan 10
name Administration
```

HR VLAN

```
vlan 20
name HR
```

Step 3 — Configure EtherChannel

Physical Ports Used

Fa0/1–2

Configuration

```
interface range fa0/1-2
channel-group 1 mode active
exit
```

Purpose

Creates:

Port-Channel 1

using:

LACP

Configure Trunk for Po1

```
interface port-channel 1
switchport mode trunk
exit
```

Purpose:

Allow VLAN Traffic

between:

SW1 ↔ CORE_SW

Step 4 — Configure Administration Ports

Actual Ports

Fa0/3–4

Configuration

```
interface range fa0/3-4
switchport mode access
switchport access vlan 10
no shutdown
exit
```

Purpose

Assigns:

Administration PCs

to:

VLAN 10

Step 5 — Configure HR Ports

Actual Ports

Fa0/5–7

Configuration

```
interface range fa0/5-7
switchport mode access
switchport access vlan 20
no shutdown
exit
```

Purpose

Assigns:

HR PCs

to:

VLAN 20

Step 6 — Save Configuration

```
copy running-config startup-config
```

CORE_SW Configuration for SW1

Ports Used

Fa0/5–6

Configuration

```
configure terminal
```

```
interface range fa0/5-6
channel-group 1 mode active
exit
```

```
interface port-channel 1
switchport trunk encapsulation dot1q
switchport mode trunk
exit
```

SW2 Configuration

Purpose

SW2 serves:

Finance Department
Management Department

Step 1 — Enter Configuration Mode

```
enable
configure terminal
```

Step 2 — Create VLANs

```
vlan 30
name Finance
```

```
vlan 40
name Management
```

Step 3 — Configure EtherChannel

Physical Ports

Fa0/1–2

Configuration

```
interface range fa0/1-2
channel-group 2 mode active
exit
```

Configure Trunk

```
interface port-channel 2
switchport mode trunk
exit
```

Step 4 — Configure Finance Ports

Actual Ports

Fa0/3–4

Configuration

```
interface range fa0/3-4
switchport mode access
switchport access vlan 30
no shutdown
exit
```

Step 5 — Configure Management Ports

Actual Ports

Fa0/5–6

Configuration

```
interface range fa0/5-6
switchport mode access
switchport access vlan 40
no shutdown
exit
```

Device Mapping

Fa0/5 → PC-M1
Fa0/6 → PC-M2

Step 6 — Save Configuration

```
copy running-config startup-config
```

CORE_SW Configuration for SW2

Ports Used

Fa0/3–4

Configuration

```
configure terminal
```

```
interface range fa0/3-4
channel-group 2 mode active
exit
```

```
interface port-channel 2
switchport trunk encapsulation dot1q
switchport mode trunk
exit
```

SW3 Configuration

Purpose

SW3 serves:

Development Department
Server Operations

Step 1 — Enter Configuration Mode

```
enable  
configure terminal
```

Step 2 — Create VLANs

```
vlan 50  
name Development
```

```
vlan 60  
name Server_Operations
```

Step 3 — Configure EtherChannel

Physical Ports

```
Fa0/1-2
```

Configuration

```
interface range fa0/1-2  
channel-group 3 mode active  
exit
```

Configure Trunk

```
interface port-channel 3  
switchport mode trunk  
exit
```

Step 4 — Configure Development Ports

Actual Ports

Fa0/3–4

Configuration

```
interface range fa0/3-4
switchport mode access
switchport access vlan 50
no shutdown
exit
```

Step 5 — Configure Server Ports

Actual Ports

Fa0/5–6

Configuration

```
interface range fa0/5-6
switchport mode access
switchport access vlan 60
no shutdown
exit
```

Step 6 — Save Configuration

```
copy running-config startup-config
```

CORE_SW Configuration for SW3

Ports Used

Fa0/7–8

Configuration

configure terminal

```
interface range fa0/7-8
channel-group 3 mode active
exit
```

```
interface port-channel 3
switchport trunk encapsulation dot1q
switchport mode trunk
exit
```

Verify EtherChannel

Run:

```
show etherchannel summary
```

Expected Output:

```
Po1(SU)
Po2(SU)
Po3(SU)
```

Meaning

S

Layer 2

U

Working / In Use

This confirms:

EtherChannel configured successfully

Save Configuration

```
copy running-config startup-config
```

END DEVICE IP CONFIGURATION + WIRELESS AP + BRANCH ROUTER + BRANCH SWITCH + EIGRP DYNAMIC ROUTING

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End Device IP Configuration

Purpose of IP Addressing

Each device inside the network must have a:

- Unique IP Address
- Correct Subnet Mask
- Proper Default Gateway

Without proper addressing:

Network communication will fail.

The enterprise network follows:

One subnet per VLAN

to maintain:

- Security
 - Department isolation
 - Better management
 - Easy troubleshooting
-

VLAN IP Addressing Table

VLAN	Department	Network	Gateway
10	Administration	192.168.10.0/24	192.168.10.1
20	HR	192.168.20.0/24	192.168.20.1
30	Finance	192.168.30.0/24	192.168.30.1
40	Management	192.168.40.0/24	192.168.40.1
50	Development	192.168.50.0/24	192.168.50.1
60	Server Operations	192.168.60.0/24	192.168.60.1
99	IT Management	192.168.99.0/24	192.168.99.1
100	Branch Office	192.168.100.0/24	192.168.100.1

SW1 — Administration & HR Devices

Administration PCs

ADMIN_PC1

Connected Port:

SW1 Fa0/3

IP Configuration:

IP Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

ADMIN_PC2

Connected Port:

SW1 Fa0/4

IP Configuration:

IP Address: 192.168.10.11

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

HR PCs

HR_PC1

Connected Port:

SW1 Fa0/5

IP Configuration:

IP Address: 192.168.20.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.1

HR_PC2

Connected Port:

SW1 Fa0/6

IP Configuration:

IP Address: 192.168.20.11

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.20.1

SW2 — Finance & Management Devices

Finance PCs

FINANCE_PC1

Connected Port:

SW2 Fa0/3

IP Configuration:

IP Address: 192.168.30.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.30.1

FINANCE_PC2

Connected Port:

SW2 Fa0/4

IP Configuration:

IP Address: 192.168.30.11

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.30.1

Management PCs

PC-M1

Connected Port:

SW2 Fa0/5

IP Configuration:

IP Address: 192.168.40.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.40.1

PC-M2

Connected Port:

SW2 Fa0/6

IP Configuration:

IP Address: 192.168.40.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.40.1

SW3 — Development & Server Devices

Development PCs

DEV_PC1

Connected Port:

SW3 Fa0/3

IP Configuration:

IP Address: 192.168.50.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.50.1

DEV_PC2

Connected Port:

SW3 Fa0/4

IP Configuration:

IP Address: 192.168.50.11

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.50.1

Server Operations

SERVER1

Connected Port:

SW3 Fa0/5

IP Configuration:

IP Address: 192.168.60.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.60.1

SERVER2

Connected Port:

SW3 Fa0/6

IP Configuration:

IP Address: 192.168.60.11

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.60.1

IT Management Device

IT_ADMIN_PC

Connected Port:

CORE_SW Fa0/2

IP Configuration:

IP Address: 192.168.99.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.99.1

Purpose:

- SSH Login
- Router Management

- Secure Administration
-

Wireless Management Configuration

Why Wireless Management?

Management staff often require:

- Mobility
- Remote access
- Wireless monitoring

To implement this:

Wireless Access Point

was used.

MGMT_AP Configuration

Connected Port:

CORE_SW Fa0/9

Assigned VLAN:

VLAN 40

Wireless Settings

Access Point Name

MGMT_AP

SSID

MGMT_WIFI

Security Mode

WPA2-PSK

Password

Cisco@123

Wireless End Devices

MGMT_LAPTOP

IP Configuration:

IP Address: 192.168.40.20

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.40.1

MGMT_PHONE

IP Configuration:

IP Address: 192.168.40.21

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.40.1

Branch Office Configuration

Purpose

The branch office must communicate with:

Head Office Network
through:

Dynamic Routing

Actual Branch Topology

```
graph TD
    MR["Main_Router Se0/3/0"] --- BR1["Branch_Router Se0/3/0"]
    BR1 --- BR2["Branch_Router Gi0/0"]
    BR2 --- BSW["Branch_SW Fa0/6"]
    BSW --- PCs["Fa0/1-5 → Branch PCs"]
```

Branch Router Configuration

Step 1 — Enter Configuration Mode

```
enable
configure terminal
```

Step 2 — Configure Hostname

```
hostname Branch_Router
```

Step 3 — Configure WAN Interface

```
interface serial0/3/0
```

```
ip address 172.16.0.2 255.255.255.252
no shutdown
exit
```

Purpose:

Main Router ↔ Branch Router communication

Step 4 — Configure LAN Interface

```
interface gigabitEthernet0/0
ip address 192.168.100.1 255.255.255.0
no shutdown
exit
```

Purpose:

Gateway for Branch PCs

Step 5 — Save Configuration

```
copy running-config startup-config
```

Branch Switch Configuration

Purpose

Provides connectivity for:

5 Branch PCs

Configure Access Ports

```
enable
```

configure terminal

interface range fa0/1-5
switchport mode access
no shutdown
exit

Configure Router Uplink

interface fa0/6
switchport mode access
no shutdown
exit

Save Configuration

copy running-config startup-config

Branch PCs Configuration

Device	IP Address
BRANCH_PC1	192.168.100.10
BRANCH_PC2	192.168.100.11
BRANCH_PC3	192.168.100.12
BRANCH_PC4	192.168.100.13
BRANCH_PC5	192.168.100.14

Subnet Mask:

255.255.255.0

Gateway:

192.168.100.1

EIGRP Dynamic Routing

Why EIGRP?

Instead of static routes:

Routes are learned automatically

Benefits:

- Fast convergence
- Enterprise friendly
- Cisco optimized

Main Router EIGRP

```
configure terminal
router eigrp 100
network 172.16.0.0 0.0.0.3
network 192.168.10.0 0.0.0.255
network 192.168.20.0 0.0.0.255
network 192.168.30.0 0.0.0.255
network 192.168.40.0 0.0.0.255
network 192.168.50.0 0.0.0.255
network 192.168.60.0 0.0.0.255
network 192.168.99.0 0.0.0.255
no auto-summary
exit
```

Branch Router EIGRP

```
configure terminal

router eigrp 100
network 172.16.0.0 0.0.0.3
network 192.168.100.0 0.0.0.255
no auto-summary
exit
```

Verify Routing

```
show ip route
```

Expected:

D 192.168.x.x

Meaning:

EIGRP learned route

ACL SECURITY + SSH + PASSWORD
ENCRYPTION + BANNER + DISABLE

ACL Security Implementation

What is ACL?

ACL (**Access Control List**) is a security mechanism used to:

- Permit traffic
- Deny traffic
- Restrict unauthorized communication
- Control network access

In this project, ACL was implemented to provide:

Department-Level Security

Why ACL Was Used

Without ACL:

All departments can communicate freely

which reduces enterprise security.

The requirement of this project was:

- Department isolation
- Controlled communication
- Management access
- Secure enterprise traffic

Therefore ACLs were applied to:

Non-management VLANs

Working ACL Logic Used in This Project

Same Department Communication

Allowed

Cross Department Communication

Restricted through ACL filtering.

Management Department

Full Access

Management users can communicate with:

- Administration
- HR
- Finance
- Development
- Server Operations

IT Management VLAN

Used for:

SSH

Remote Management

ACL Configuration on Main Router

ACLs were configured on:

Router Subinterfaces

because Inter-VLAN Routing occurs on the router.

ACL for VLAN 10 — Administration

Purpose

Restricts Administration users from accessing unauthorized departments.

Configuration

```
access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255
access-list 110 deny ip 192.168.10.0 0.0.0.255 192.168.60.0 0.0.0.255
access-list 110 permit ip any any
```

Apply ACL

```
interface gigabitEthernet0/1.10
ip access-group 110 in
exit
```

ACL for VLAN 20 — HR

Configuration

```
access-list 120 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 120 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 120 deny ip 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255
access-list 120 deny ip 192.168.20.0 0.0.0.255 192.168.60.0 0.0.0.255
access-list 120 permit ip any any
```

Apply ACL

```
interface gigabitEthernet0/1.20
ip access-group 120 in
exit
```

ACL for VLAN 30 — Finance

Configuration

```
access-list 130 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 130 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 130 deny ip 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255
access-list 130 deny ip 192.168.30.0 0.0.0.255 192.168.60.0 0.0.0.255
access-list 130 permit ip any any
```

Apply ACL

```
interface gigabitEthernet0/1.30
ip access-group 130 in
exit
```

ACL for VLAN 50 — Development

Configuration

```
access-list 150 deny ip 192.168.50.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 150 deny ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 150 deny ip 192.168.50.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 150 deny ip 192.168.50.0 0.0.0.255 192.168.60.0 0.0.0.255
access-list 150 permit ip any any
```

Apply ACL

```
interface gigabitEthernet0/1.50
ip access-group 150 in
exit
```

Why VLAN 40 Has No ACL?

Management VLAN:

192.168.40.0/24

was intentionally left unrestricted because:

Management requires full enterprise access

This fulfills the project requirement.

SSH Secure Remote Management

Why SSH?

SSH provides:

Encrypted Remote Access

Unlike:

Telnet

which sends credentials in plain text.

Main Router SSH Configuration

configure terminal

hostname Main_Router

ip domain-name technova.local

username admin privilege 15 secret Cisco@123

Generate RSA Key

crypto key generate rsa

When prompted:

1024

Configure VTY Lines

```
line vty 0 4
transport input ssh
login local
exit
```

```
ip ssh version 2
```

CORE_SW SSH Configuration

```
configure terminal
```

```
hostname CORE_SW
ip domain-name technova.local
```

```
username admin privilege 15 secret Cisco@123
```

Generate RSA Key

```
crypto key generate rsa
```

Type:

```
1024
```

Configure VLAN 99 Interface

```
interface vlan 99
ip address 192.168.99.2 255.255.255.0
no shutdown
exit
```

Configure Default Gateway

```
ip default-gateway 192.168.99.1
```

Configure SSH Access

```
line vty 0 4
transport input ssh
login local
exit
```

```
ip ssh version 2
```

Password Encryption

Purpose

Protects passwords from being displayed as plain text.

Commands

```
service password-encryption
enable secret Cisco@123
```

Login Banner (MOTD)

Purpose

Displays security warning during login.

Command

```
banner motd # WARNING: Unauthorized access is prohibited! #
```

Disable Unused Ports

Why?

Unused switch ports may become:

Security Vulnerabilities

Therefore unused ports were disabled.

SW1

```
interface range fa0/7-24
shutdown
exit
```

SW2

```
interface range fa0/7-24
shutdown
exit
```

SW3

```
interface range fa0/7-24
shutdown
exit
```


Final Verification Commands

Verify Interfaces

show ip interface brief

Expected:

up/up

Verify Trunking

show interfaces trunk

Expected:

Fa0/1 trunking

Po1 trunking

Po2 trunking

Po3 trunking

Verify EtherChannel

show etherchannel summary

Expected:

Po1(SU)

Po2(SU)

Po3(SU)

Verify Routing

show ip route

Expected:

D 192.168.x.x

Verify SSH

From:

IT_ADMIN_PC

Run:

ssh -l admin 192.168.99.1

Password:

Cisco@123

Expected:

Main_Router#

Troubleshooting Guide

VLAN Not Working

Check:

show vlan brief

Trunk Problem

Check:

show interfaces trunk

EtherChannel Problem

Check:

show etherchannel summary

Correct State:

SU

SSH Problem

Check:

- VLAN 99
- IT_ADMIN_PC IP
- SSH configuration
- Router reachability

Test:

ping 192.168.99.1

**END OF FULL PROJECT
DOCUMENTATION** 